

Bachelor of Technology — General Engineering - Open Specialisation

Table 1: General Engineering - Open Specialisation — B25 Credit Distribution

Division	Subdivision	Credits
IC Core Courses	IC Compulsory	39
	IC Basket	6
	IKS + HSS	15
Discipline Courses	Discipline Core	35
	Discipline Electives (merged into Free Electives)	—
Electives	Free Electives (including merged DE pool)	53
	MTP (4+4, equivalent as DE)	8
	ISTP (equivalent as FE)	4
Total		160

Table 2: B. Tech General Engineering - Open Specialisation — B25 Discipline Core Courses

Sl. No	Course Code	Course Name	L-T-P-C	Credits
1	ME-100	Reverse Engineering	0-0-2-1	1
2	EE-261	Electrical System Around Us	3-0-0-3	3
3	EE-231	Measurement and Instrumentation	2-0-2-3	3
4	ME-213	Thermodynamics	3-1-0-4	4
5	EE-201	Electromechanics	2.5-0.5-0-3	3
6	EE-211	Analog Circuit Design	2-0-2-3	3
7	ME-212	Product Manufacturing Technology	2-0-3-3	3
8	IC-241	Material Science for Engineers	3-0-0-3	3
9	IC-253	Programming and Data Structures	2-0-2-3	3
10	HS-307	Macroeconomics	3-0-0-3	3
11	HS-541	Technical Communication	1-0-0-1	1
12	ME-206	Mechanics of Solids	2.5-0.5-0-3	3
Total				33

Table 3: B. Tech General Engineering - Open Specialisation — B25 Semester 1

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-112	Calculus	1.5-0.5-0-2	2
2	IC	IC-114	Linear Algebra	2-0-0-2	2
3	IC	IC-152	Computing and Data Science	3-0-2-4	4
4	IC Basket	IC-230	Environmental Science	3-0-0-3	3
5	IKS + HSS (choose 1)	HS-108	Basic English for Engineers	3-0-0-3	3
6		HS-112	Japanese Language Class I	3-0-0-3	
7		HS-256	Introduction to Spanish	3	
8		HS-342	German I	3-0-0-3	
9	IKS + HSS (choose 1)	IC-181	Indian Knowledge Systems	2-0-2-3	3
10		IC-182	History of Science and Technology	3	
Total					17

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 4: B. Tech General Engineering - Open Specialisation — B25 Semester 2

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-102P	Foundations of Design Practicum	1-0-6-4	4
2	IC	IC-113	Complex Variables and Vector Calculus	1.5-0.5-0-2	2
3	IC	IC-115	ODE & Integral Transforms	2-0-0-2	2
4	IC	IC-140	Engineering Graphics for Design	2-0-3-4	4
5	IC	IC-161	Applied Electronics	3	3
6	IC	IC-161P	Applied Electronics Lab	2	2
7	IC	IC-252	Probability and Statistics	3-0-2-4	4
8	IC Basket	IC-240	Mechanics of Rigid Bodies	1.5-1.5-0-3	3
Total					24

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 5: B. Tech General Engineering - Open Specialisation — B25 Semester 3

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-202P	Design Practicum	0-0-6-3	3
2	IC	IC-272	Machine Learning	2-0-2-3	3
3	DC	EE-261	Electrical System Around Us	3-0-0-3	3
4	DC	IC-241	Material Science for Engineers	3-0-0-3	3
5	DC	ME-212	Product Manufacturing Technology	2-0-3-3	3
6	DC	ME-206	Mechanics of Solids	2.5-0.5-0-3	3
Total					18

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 6: B. Tech General Engineering - Open Specialisation — B25 Semester 4

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	EE-201	Electromechanics	2.5-0.5-0-3	3
2	DC	EE-211	Analog Circuit Design	2-0-2-3	3
3	DC	HS-307	Macroeconomics	3-0-0-3	3
4	DC	IC-253	Programming and Data Structures	2-0-2-3	3
5	DC	ME-100	Reverse Engineering	0-0-2-1	1
Total					13

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 7: B. Tech General Engineering - Open Specialisation — B25 Semester 5

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	EE-231	Measurement and Instrumentation	2-0-2-3	3
2	DC	ME-213	Thermodynamics	3-1-0-4	4
Total					7

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 8: B. Tech General Engineering - Open Specialisation — B25 Semester 6

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	HS-541	Technical Communication	1-0-0-1	1
2	ISTP	DP-301P	Interdisciplinary Socio-Technical Practicum	0-0-6-4	4
Total					5

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 9: B. Tech General Engineering - Open Specialisation — B25 Semester 7

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	MTP	GE-498P	Major Technical Project - I	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 10: B. Tech General Engineering - Open Specialisation — B25 Semester 8

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-010	Internship		2
2	MTP	GE-499P	Major Technical Project - II	0-0-6-4	4
Total					6

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

*Beyond the core courses above, students choose 53 credits of **Free Electives** — for this major the Discipline Elective and Free Elective pools are merged into a single pool — from across the institute, subject to the elective policy below.*

Elective Policy. There is no prescribed Discipline Elective list for this major. Students may choose electives from any department, provided that no more than 40% of their elective credits come from the Discipline Core list of any single named major. The Discipline Elective and Free Elective pools are merged into one pool of 53 credits of Free Electives, giving students maximum flexibility in course selection.

The following compulsory project components count toward the DE credit requirement; they are not selectable DE options.

Table 11: General Engineering - Open Specialisation — Compulsory project components credited as DE

Sl. No	Course Code	Course Name	Credits
1	GE-498P	Major Technical Project - I (compulsory; counted as DE)	4
2	GE-499P	Major Technical Project - II (compulsory; counted as DE)	4